UC3M



PACKAGE REFERENCE DOCUMENT

Contents

1 Code display packages **3**
1.1 Listings – Source Code . . . 3
1.2 Xcolor – Colors . . . 3

2 Graphics packages **3**
2.1 TikZ – Block Diagrams . . . 3
2.1.1 Basic Block Diagram . . . 4
2.1.2 Node Positioning . . . 4
2.1.3 Node Types . . . 4
2.1.4 Connections . . . 4
2.1.5 Summing Junctions . . . 5
2.1.6 Pickoff Points . . . 5
2.1.7 Feedback Loops . . . 5
2.1.8 Useful Coordinates . . . 6
2.1.9 Scaling . . . 6
2.2 Images . . . 6
2.2.1 Single Image . . . 6
2.2.2 Image Size . . . 7
2.2.3 Two Images Side by Side . . . 7
2.2.4 Image Wrapped by Text . . . 8
2.2.5 Figure Positioning . . . 9
2.2.6 Captions and References . . . 9

3 Document layout, page layout and formatting packages **9**
3.1 Internal and External Links . . . 9
3.2 Needspace – Page Break Control . . . 10
3.2.1 Basic Usage . . . 10
3.2.2 Required Space . . . 10
3.2.3 Headings and Following Text . . . 11
3.2.4 Short Lists . . . 11
3.2.5 Command Variants . . . 11
3.2.6 Practical Limitations . . . 12
3.3 Multicol – Multiple Columns . . . 12

4 Mathematical typesetting packages **13**
4.1 Cancel – Cancellations . . . 13
4.1.1 Basic Cancellation . . . 13
4.1.2 Cancelling Common Terms . . . 13
4.1.3 Cancelling to a Value . . . 13
4.1.4 Other Cancellation Styles . . . 13
4.2 Amsmath – Mathematical Expressions . . . 14
4.2.1 Equations . . . 14
4.2.2 Aligned Equations . . . 14
4.2.3 Multiple Lines in One Equation . . . 15
4.2.4 Matrices . . . 15
4.3 Siunitx – Numbers and Units . . . 15
4.3.1 Physical Quantities . . . 16
4.3.2 Units Without Values . . . 16
4.3.3 Numbers . . . 16

4.3.4 Angles 16

5 Text formatting packages **17**

1 Code display packages

1.1 Listings – Source Code

The `listings` package is used to display source code while preserving indentation and formatting. Different styles can be used depending on the programming language.

For example, MATLAB code can be displayed using the `matlab` style:

```
clc
clear

x = linspace(0, 2*pi, 100);
y = sin(x);

plot(x,y)
```

The corresponding environment is opened with:

```
\begin{lstlisting}[style=matlab]
```

and closed using `\end{lstlisting}`.

The appearance of the code listings is configured in `Packages/Code/package_config.tex`. This file defines the available styles, including properties such as the font, colors, background, line numbering and spacing.

When adding or modifying a code style, it is recommended to make the changes in this configuration file rather than changing individual listings throughout the document.

1.2 Xcolor – Colors

The `xcolor` package provides color support in LaTeX. In this template, it is mainly used by `listings` to add syntax highlighting and other colors to source code.

For example, a code style may define different colors for comments, keywords and strings:

```
\definecolor{codegreen}{RGB}{0,120,0}
\definecolor{codeblue}{RGB}{0,70,180}

\lstdefinestyle{example}{
    keywordstyle=\color{codeblue},
    commentstyle=\color{codegreen}
}
```

Normally, these settings do not need to be defined inside the main document. The code colors and listing styles are configured in `Packages/Code/package_config.tex`.

2 Graphics packages

2.1 TikZ – Block Diagrams

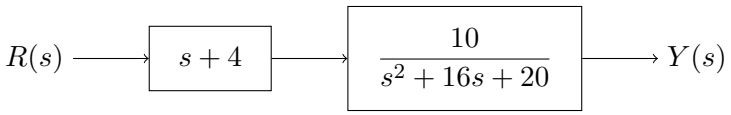
For control-system block diagrams, TikZ requires:

```
\usepackage{tikz}
\usetikzlibrary{positioning, calc}
```

2.1.1 Basic Block Diagram

```
\begin{tikzpicture}
    \node[input] (input) {$R(s)$};
    \node[block, right=of input] (G1) {$s+4$};
    \node[block, right=of G1] (G2)
        {$\dfrac{10}{s^2+16s+20}$};
    \node[output, right=of G2] (output) {$Y(s)$};

    \draw[->] (input) -- (G1);
    \draw[->] (G1) -- (G2);
    \draw[->] (G2) -- (output);
\end{tikzpicture}
```



2.1.2 Node Positioning

```
right=of node
left=of node
above=of node
below=of node

right=1cm of node
below=2\dnd cm of node % 2 times the default node distance
```

The first form uses the default `node distance`; the second overrides it locally.

2.1.3 Node Types

```
\node[input]   (input) {$R(s)$};

\node[block]   (G) {$G(s)$};

\node[sum]     (sum1) {};

\node[pickoff] (p1) {};

\node[output]  (output) {$Y(s)$};
```

2.1.4 Connections

```
% Straight line
\draw[->] (A) -- (B);

% Straight line without arrow
\draw[-] (A) -- (B);

% Vertical, then horizontal
```

```
\draw[->] (A) |- (B);

% Horizontal, then vertical
\draw[->] (A) -| (B);
```

Node anchors can be selected explicitly:

```
node.north
node.south
node.east
node.west
```

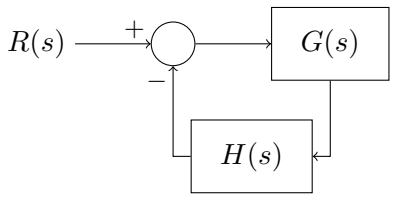
2.1.5 Summing Junctions

Like a connection, but with a plus or minus sign at the junction. The following code snippet will be used after positioning the nodes:

```
\node[sum] (sum1) {};

\draw[->] (G)
    -- node[plus, above left] {}
    (sum1);

\draw[->] (H)
    -| node[minus, below left] {}
    (sum1.south);
```



2.1.6 Pickoff Points

```
\node[pickoff, right=of G] (p1) {};

\draw[-]  (G) -- (p1);
\draw[->] (p1) -- (output);
\draw[->] (p1.south) |- (feedback.east);
```

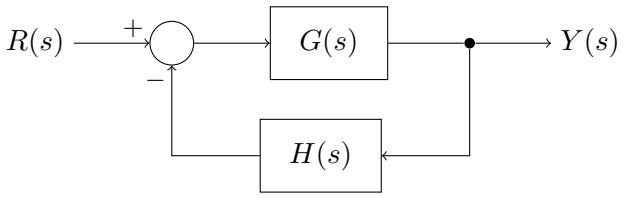
2.1.7 Feedback Loops

```
\coordinate (mid)
    at ($(sum1)!0.5!(pickoff1)$);

\node[block, below=of mid] (feedback)
    {$H(s)$};

\draw[->] (pickoff1.south)
    |- (feedback.east);
```

```
\draw[->] (feedback.west)
    -| node[minus, below left] {}
    (sum1.south);
```



2.1.8 Useful Coordinates

Point halfway between two nodes:

```
\coordinate (mid)
    at ($(A)!0.5!(B)$);
```

Other positions:

```
($(A)!0.25!(B)$)   % 25 %
($(A)!0.50!(B)$)   % 50 %
($(A)!0.75!(B)$)   % 75 %
```

Where A and B are the names of the nodes. The `calc` library must be loaded for this to work.

2.1.9 Scaling

In case the diagram doesn't fit in the page, we can scale it down using the `scale` option in the `tikzpicture` environment:

```
\begin{tikzpicture}[scale=0.8, transform shape]
    ...
\end{tikzpicture}
```

2.2 Images

Images are inserted using the following packages:

```
\usepackage{graphicx}   % Images
\usepackage{float}      % [H] positioning
\usepackage{wrapfig}    % Wrapped figures
```

2.2.1 Single Image

```
\begin{figure}[H]
    \centering
    \includegraphics[width=0.8\linewidth]
        {Images/image_placeholder.png}
    \caption{Image caption}
    \label{fig:image}
```

```
\end{figure}
```

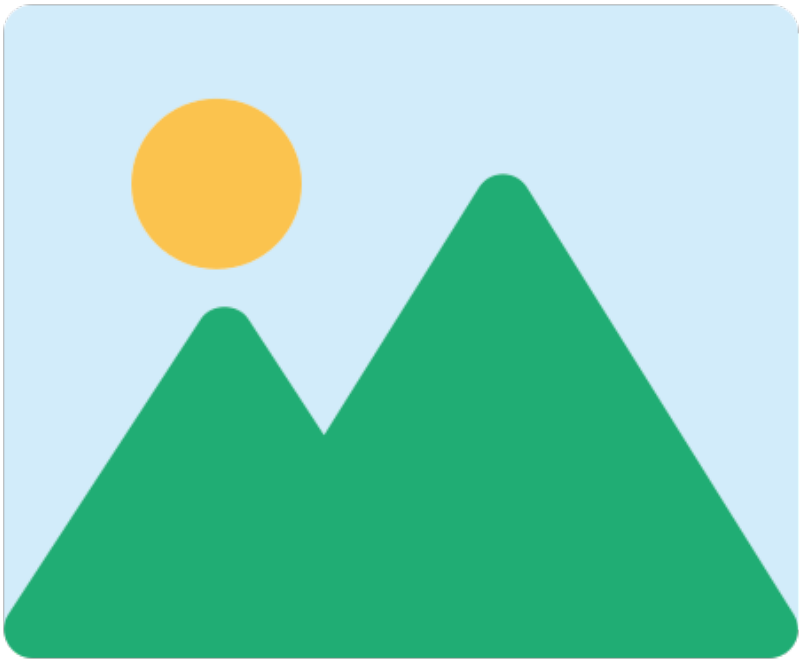


Figure 1: Image caption

2.2.2 Image Size

The image size can be controlled using `width`:

```
\includegraphics[width=\linewidth]
    {Images/image_placeholder.png}

\includegraphics[width=0.8\linewidth]
    {Images/image_placeholder.png}

\includegraphics[width=0.5\textwidth]
    {Images/image_placeholder.png}
```

It can also be controlled using its height:

```
\includegraphics[height=5cm]
    {Images/image_placeholder.png}
```

2.2.3 Two Images Side by Side

```
\begin{figure}[H]
    \centering

    \begin{minipage}{0.47\textwidth}
        \centering
        \includegraphics[width=\linewidth]
            {Images/image_placeholder.png}
        \caption{First image}
        \label{fig:image_1}
    \end{minipage}
    \hfill
    \begin{minipage}{0.47\textwidth}
        \centering
        \includegraphics[width=\linewidth]
            {Images/image_placeholder.png}
        \caption{Second image}
        \label{fig:image_2}
    \end{minipage}

\end{figure}
```

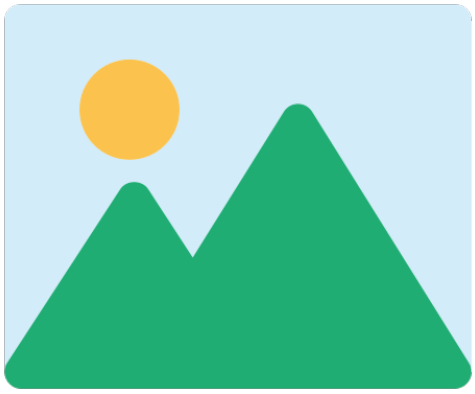


Figure 2: First image

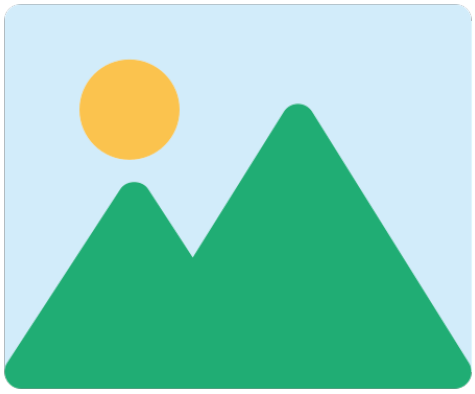


Figure 3: Second image

2.2.4 Image Wrapped by Text

Use `wrapfigure` to place an image beside the text.

```
\begin{wrapfigure}[8]{r}{0.42\textwidth}
    \centering
    \includegraphics[width=0.40\textwidth]
        {Images/image_placeholder.png}
    \caption{Image caption}
    \label{fig:wrapped_image}
\end{wrapfigure}
```

This text flows around the image. The first argument of `wrapfigure` determines its position. For example, `r` places the image on the right and `l` places it on the left. The number between brackets represents the number of lines wrapping around the image.

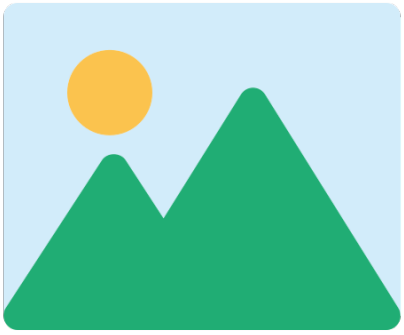


Figure 4: Wrapped image

2.2.5 Figure Positioning

Common figure positioning options are:

```
\begin{figure}[H]     % Exactly here
\begin{figure}[h]     % Preferably here
\begin{figure}[t]     % Top of page
\begin{figure}[b]     % Bottom of page
\begin{figure}[htbp] % Let LaTeX choose
```

Using `[H]` requires the `float` package.

2.2.6 Captions and References

Add a label immediately after the caption:

```
\caption{Image caption}
\label{fig:example}
```

The figure can then be referenced in the text:

```
Figure~\ref{fig:example}
```

3 Document layout, page layout and formatting packages

3.1 Internal and External Links

The `hyperref` package allows clickable links inside the document, including references, the table of contents, URLs, and links to other sections.

```
\usepackage{hyperref}

\hypersetup{
    colorlinks=true,
    linkcolor=blue,       % Internal links
    filecolor=magenta,
    urlcolor=cyan,
    linktoc=all           % Title and page number are clickable
}
```

To link to another part of the document, first create a label:

```
\section{Control Systems}
\label{sec:control}
```

Then reference it using:

```
See Section~\ref{sec:control}.
```

A clickable text link can also be created with:

```
\hyperref[sec:control]{Go to Control Systems}
```

For external URLs:

```
\url{https://example.com}

\href{https://example.com}{Example website}
```

If visible link colours or borders are not desired, use:

```
\usepackage[hidelinks]{hyperref}
```

With `hidelinks`, the links remain clickable, but no coloured text or link borders are shown.

3.2 Needspace – Page Break Control

The `needspace` package starts a new page when there is not enough space for the material that follows. Load it in the preamble:

```
\usepackage{needspace}
```

3.2.1 Basic Usage

Place `\Needspace` between paragraphs, immediately before the content that requires a minimum amount of space:

```
Previous paragraph.

\Needspace{5\baselineskip}
This paragraph starts on the next page if less than
five baseline spacings remain on the current page.
```

If enough space remains, the text continues normally. The command does not insert an empty gap.

3.2.2 Required Space

The argument is a vertical length. It can be expressed using physical units, the current line spacing or the text height:

```
% Alternative ways to specify the required space:
\Needspace{3cm}
\Needspace{5\baselineskip}
```

```
\Needspace{0.25\textheight}
```

Here, `\baselineskip` is the distance between consecutive text baselines. The last example requests one quarter of the text area's height.

3.2.3 Headings and Following Text

Place the command before a heading to leave room for both the heading and the beginning of its text:

```
\Needspace{7\baselineskip}
\subsection{Results}

The following results describe the response of the
system under the selected operating conditions.
```

The requested space should include the heading, its vertical spacing and the desired number of text lines.

3.2.4 Short Lists

Reserve space before the introductory sentence to avoid separating it from the beginning of a list:

```
\Needspace{8\baselineskip}
The main assumptions are:

\begin{itemize}
    \item Steady flow.
    \item Constant fluid density.
    \item Negligible viscous losses.
\end{itemize}
```

To keep the entire list together, allow enough space for all items and the spacing around them. Increase the requested length if the items occupy multiple lines.

3.2.5 Command Variants

The package provides an approximate command and two variants that check the available space directly:

```
% Approximate space reservation
\needspace{5\baselineskip}

% Exact space check
\Needspace{5\baselineskip}

% Exact space check; preserves bottom alignment
\Needspace*{5\baselineskip}
```

The lowercase command is faster but less precise. Both capitalized variants should be used between paragraphs.

When a page is ended early, `\needspace` and `\Needspace` leave unused space at the bottom. The starred variant respects `\flushbottom` when that setting is active:

```
% Document setting
\flushbottom

% Later, between paragraphs
\Needspace*{6\baselineskip}
The following material requires additional room.
```

With `\flushbottom`, stretchable vertical spacing can expand to align the bottom of the page. The star does not enable `\flushbottom` by itself.

3.2.6 Practical Limitations

The package checks the requested length, not the actual height of the following content. A paragraph or list that exceeds that length can still continue on another page.

Use the command in the main text flow, before the relevant material. It does not control the placement of floating figures or tables. In a two-column layout, an insufficient amount of space causes a break to the next column.

3.3 Multicol – Multiple Columns

The `multicol` package divides part of the document into multiple columns. The number of columns is specified when opening the environment:

```
\begin{multicols*}{4}

    Content of the first column

    \columnbreak

    Content of the second column

    \columnbreak

    Content of the third column

    \columnbreak

    Content of the fourth column

\end{multicols*}
```

The example above creates four columns. The `\columnbreak` command can be used to manually move to the next column.

This is particularly useful when several short sections must be arranged compactly on the same page:

```
\begin{multicols*}{3}
    \input{sections/hooke}
    \input{sections/bending}

    \columnbreak

    \input{sections/torsion}
    \input{sections/power}
```

```
    \columnbreak

    \input{sections/buckling}
\end{multicols*}
```

The starred environment `multicols*` does not automatically balance the height of the final columns. This is useful when the position of the content is controlled manually using `\columnbreak`.

4 Mathematical typesetting packages

4.1 Cancel – Cancellations

The `cancel` package is used to visually cancel terms in mathematical expressions.

4.1.1 Basic Cancellation

```
\[
\cancel{x}
\]
```

$$\cancel{x}$$

4.1.2 Cancelling Common Terms

It can be used to show the simplification of common terms:

```
\[
\frac{\cancel{x}y}{\cancel{x}z}
=
\frac{y}{z}
\]
```

$$\frac{\cancel{x}y}{\cancel{x}z} = \frac{y}{z}$$

4.1.3 Cancelling to a Value

The `\cancelto` command indicates the value to which a term is reduced:

```
\[
a + \cancelto{0}{b-b} = a
\]
```

$$a + \cancelto{0}{b-b} = a$$

4.1.4 Other Cancellation Styles

Different cancellation directions are also available:

```
\[
\cancel{x}
\qquad
\bcancel{x}
\qquad
\xcancel{x}
\]
```

$$\cancel{x} \qquad \bcancel{x} \qquad \xcancel{x}$$

4.2 Amsmath – Mathematical Expressions

The `amsmath` package extends the mathematical capabilities of LaTeX, providing additional environments for equations, alignment, and matrices.

4.2.1 Equations

A numbered equation can be created using the `equation` environment:

```
\begin{equation}
    F = ma
\end{equation}
```

$$F = ma \tag{1}$$

The starred version removes the equation number:

```
\begin{equation*}
    E = mc^2
\end{equation*}
```

$$E = mc^2$$

4.2.2 Aligned Equations

The `align` environment aligns several equations at a common point. The alignment point is indicated using `&`:

```
\begin{align}
    F &= ma \\
    a &= \frac{F}{m}
\end{align}
```

$$F = ma \tag{2}$$

$$a = \frac{F}{m} \tag{3}$$

4.2.3 Multiple Lines in One Equation

The `split` environment can be used when a single equation must be written across several lines:

```
\begin{equation}
\begin{split}
    E &= T + V \\
      &= \frac{1}{2}mv^2 + mgh
\end{split}
\end{equation}
```

$$\begin{aligned} E &= T + V \\ &= \frac{1}{2}mv^2 + mgh \end{aligned} \tag{4}$$

All lines share the same equation number.

4.2.4 Matrices

Different matrix environments can be used depending on the desired delimiters:

```
\[
A =
\begin{bmatrix}
    1 & 2 \\
    3 & 4
\end{bmatrix}
\]
```

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Other common matrix types are:

```
\[
\begin{pmatrix}
    1 & 2 \\
    3 & 4
\end{pmatrix}
\qquad
\begin{vmatrix}
    1 & 2 \\
    3 & 4
\end{vmatrix}
\]
```

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \qquad \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$$

4.3 Siunitx – Numbers and Units

The `siunitx` package provides consistent formatting for numbers, physical quantities, and units.

4.3.1 Physical Quantities

Use `\qty` to write a numerical value together with its unit:

```
\[
v = \qty{25}{\meter\per\second}
\]
```

$$v = 25\,\mathrm{m\,s^{-1}}$$

More complex units can be constructed by combining unit commands:

```
\[
a = \qty{9.81}{\meter\per\second\squared}
\]

\[
\rho = \qty{1.225}{\kilogram\per\meter\cubed}
\]
```

$$a = 9.81\,\mathrm{m\,s^{-2}}$$

$$\rho = 1.225\,\mathrm{kg\,m^{-3}}$$

4.3.2 Units Without Values

The `\unit` command displays a unit without an associated numerical value:

```
\[
\unit{\meter\per\second}
\qquad
\unit{\newton}
\qquad
\unit{\pascal}
\]
```

$$\mathrm{m\,s^{-1}} \qquad \mathrm{N} \qquad \mathrm{Pa}$$

4.3.3 Numbers

The `\num` command provides consistent formatting for numerical values:

```
\num{12500}

\num{3.2e8}
```

$$12\,500 \qquad 3.2 \times 10^{8}$$

4.3.4 Angles

Angles can be written using `\ang`:

```
\[
\alpha = \ang{30}
\qquad
\beta = \ang{45}
\qquad
\gamma = \ang{90}
\]
```

$$\alpha = 30^\circ \qquad \beta = 45^\circ \qquad \gamma = 90^\circ$$

5 Text formatting packages